

ASSIGNMENT-1

1	In a metal shop two articles are produced, i.e., bucket lids and pressed metal handles. The lid takes 20 seconds to stamp, 30 secs. to form and 20 secs. to paint. The handle takes 40 secs. to stamp, 10 secs. to form and 60 secs. to paint. The profit margin on lids and handles are 6 and 9 paise respectively. The time available per day on each process is 8 hours and 20 minutes. (i) How many lids and handles should be produced to maximize profit? Find also the maximum profit. (ii) Will the production policy change if profit margins were reversed? How much profit will be earned then? (iii) What will be the production policy and maximum gross profit with given original objective function and capacities, if there is a market restriction of 750 handles?
2	Find basic solutions; identify basic feasible solutions, degenerate basic feasible solutions. Show that there is a correspondence with the graphical corner points: $\text{Max} Z = 4x_1 + 2x_2$ subject to constraints $x_1 + x_2 + x_3 = 3$ $x_1 - x_2 + x_4 = 0$ $x_1, x_2, x_3, x_4 \geq 0.$
3	Solve the following LPP: Maximize: $Z = 4x_1 + x_2 + 3x_3 + 5x_4$ subject to the constraints: $4x_1 - 6x_2 - 5x_3 - 4x_4 \geq -20,$ $-3x_1 - 2x_2 + 4x_3 + x_4 \leq 10$ $-8x_1 - 3x_2 + 3x_3 + 2x_4 \leq 20$ and $x_1, x_2, x_3, x_4 \geq 0.$
4	Solve the following LPP: Max. $Z = x_1 + 3x_2$ subject to the constraints: $x_1 + x_2 \leq 2,$ $-x_1 + x_2 \leq 4,$ and $x_2 \geq 0, x_1$ is unrestricted
5	Use penalty (Big-M) method to Maximize: $Z = 3x_1 - x_2$ subject to the constraints: $2x_1 + x_2 \geq 2,$ $x_1 + 3x_2 \leq 3,$ $x_2 \leq 4$ and $x_1, x_2 \geq 0.$
6	Solve by using big-M method the following LP problem: Min. $Z = -x_1 - 2x_2 - 3x_3 + x_4 - 25$

	subject to the constraints $x_1 + 2x_2 + 3x_3 = 15$ $2x_1 + x_2 + 5x_3 = 20$ $x_1 + 2x_2 + x_3 + x_4 = 10$ and $x_1, x_2, x_3, x_4 \geq 0$.																																																								
7	Use the two-phase simplex method to Max. $Z = 2x_1 + x_2 + \frac{1}{4}x_3$ subject to the constraints $4x_1 + 6x_2 + 3x_3 \leq 8$ $3x_1 - 6x_2 - 3x_3 \leq 1$ $2x_1 + 3x_2 - 5x_3 \geq 4$ and $x_1, x_2, x_3 \geq 0$.																																																								
8	Use the two-phase simplex method to solve the following LPP Min. $Z = x_1 - x_2$ subject to the constraints: $x_1 + x_2 \leq 1,$ $2x_1 + 3x_2 \geq 6,$ and $x_1, x_2 \geq 0$.																																																								
9	Solve the following LPP: Max. $Z = 5x_1 + 3x_2$ subject to the constraints: $x_1 + x_2 \leq 2,$ $5x_1 + 2x_2 \leq 10,$ $3x_1 + 8x_2 \leq 12$ and $x_1, x_2 \geq 0$.																																																								
10	The two consecutive simplex tables of an LPP are given as follows: <table><tr><th>Basic Variable</th><th>C_B</th><th>X_B</th><th>x_1</th><th>x_2</th><th>x_3</th><th>x_4</th><th>x_5</th></tr><tr><td>x_4</td><td>-</td><td>9</td><td>A</td><td>B</td><td>C</td><td>D</td><td>0</td></tr><tr><td>x_5</td><td>-</td><td>2</td><td>-2</td><td>-1</td><td>E</td><td>0</td><td>F</td></tr><tr><td></td><td>$Z_j - C_j$</td><td></td><td>G</td><td>-1</td><td>3</td><td>0</td><td>0</td></tr><tr><td>x_1</td><td>-</td><td>N</td><td>J</td><td>$\frac{2}{3}$</td><td>$\frac{2}{3}$</td><td>$\frac{1}{3}$</td><td>0</td></tr><tr><td>x_5</td><td>-</td><td>8</td><td>K</td><td>$\frac{1}{3}$</td><td>$\frac{1}{3}$</td><td>$\frac{2}{3}$</td><td>1</td></tr><tr><td></td><td>$Z_j - C_j$</td><td></td><td>0</td><td>-5</td><td>L</td><td>M</td><td>0</td></tr></table> Find the missing entries A,B,C,D,E F,G,H,J,K,L,M,N in the above table.	Basic Variable	C_B	X_B	x_1	x_2	x_3	x_4	x_5	x_4	-	9	A	B	C	D	0	x_5	-	2	-2	-1	E	0	F		$Z_j - C_j$		G	-1	3	0	0	x_1	-	N	J	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{1}{3}$	0	x_5	-	8	K	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{2}{3}$	1		$Z_j - C_j$		0	-5	L	M	0
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x_5	-	8	K	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{2}{3}$	1																																																		
	$Z_j - C_j$		0	-5	L	M	0																																																		